

firing threshold. This diminishes the generation of action potentials. Figure 8.3 shows an example of an inhibitory pathway.

C. Combined effects of the EPSP and IPSP

Most neurons in the CNS receive both EPSP and IPSP input. Thus, several different types of neurotransmitters may act on the same neuron, but each binds to its own specific receptor. The overall action is the summation of the individual actions of the various neurotransmitters on the neuron. The neurotransmitters are not uniformly distributed in the CNS but are localized in specific clusters of neurons, the axons of which may synapse with specific regions of the brain. Many neuronal tracts, thus, seem to be chemically coded, and this may offer greater opportunity for selective pharmacological modulation of certain neuronal pathways.

IV. NEURODEGENERATIVE DISEASES

Neurodegenerative diseases of the CNS include Parkinson disease, Alzheimer disease, MS, and ALS. These devastating illnesses are characterized by the progressive loss of selected neurons in discrete brain areas, resulting in characteristic disorders of movement, cognition, or both.

V. OVERVIEW OF PARKINSON DISEASE

Parkinsonism is a progressive neurological disorder of muscle movement, characterized by tremors, muscular rigidity, bradykinesia, and postural and gait abnormalities. Most cases involve people over the age of 65, among whom the incidence is about 1 in 100 individuals.

A. Etiology

The cause of Parkinson disease is unknown for most patients. The disease is correlated with destruction of dopaminergic neurons in the substantia nigra with a consequent reduction of dopamine actions in the corpus striatum, parts of the basal ganglia system that are involved in motor control.

1. Substantia nigra: The substantia nigra, part of the extrapyramidal system, is the source of dopaminergic neurons (shown in *red* in Figure 8.4) that terminate in the neostriatum. Each dopaminergic neuron makes thousands of synaptic contacts within the neostriatum and therefore modulates the activity of a large number of cells. These dopaminergic projections from the substantia nigra fire tonically rather than in response to specific muscular movements or sensory input. Thus, the dopaminergic system appears to serve as a tonic, sustaining influence on motor activity, rather than participating in specific movements.

2. Neostriatum: Normally, the neostriatum is connected to the substantia nigra by neurons (shown in *orange* in Figure 8.4) that secrete the inhibitory transmitter GABA at their termini. In turn, cells of the substantia nigra send neurons back to the neostriatum, secreting the inhibitory transmitter dopamine at their termini. This mutual inhibitory pathway normally maintains a degree of inhibition

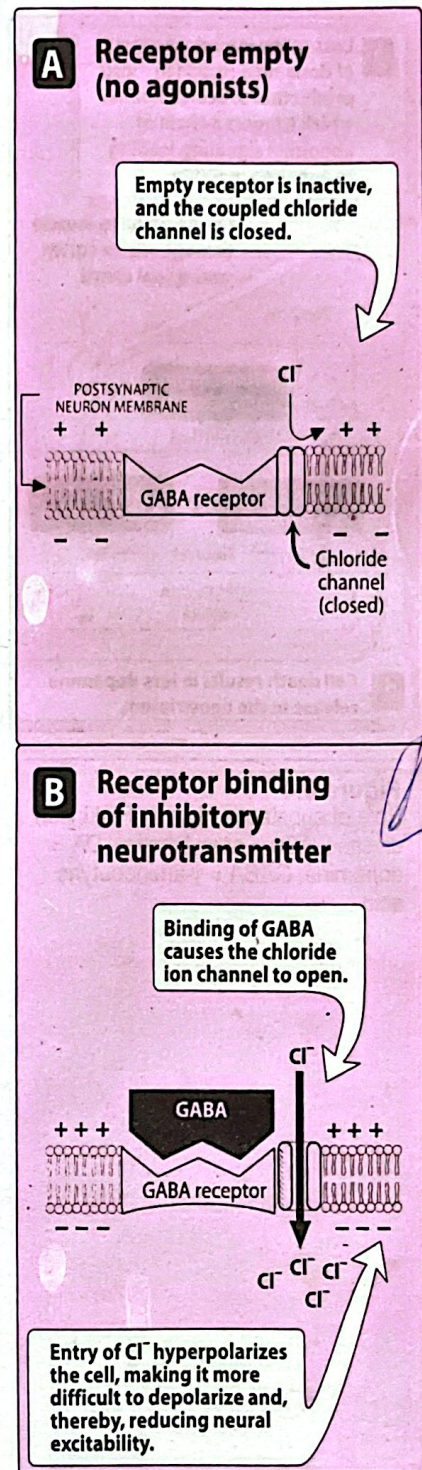
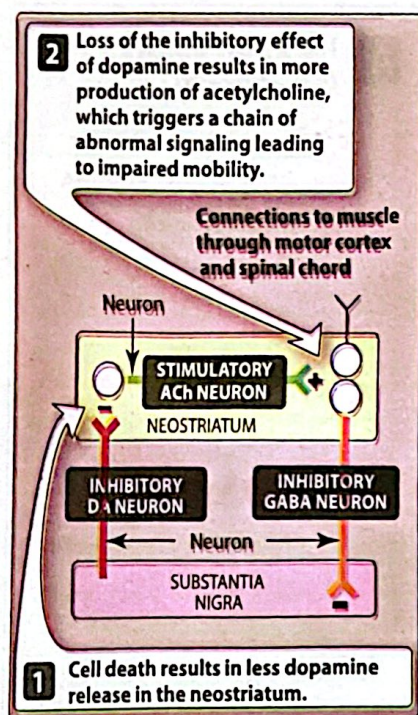


Figure 8.3

Binding of the inhibitory neurotransmitter, γ -aminobutyric acid (GABA), causes hyperpolarization of the neuron.

**Figure 8.4**

Role of substantia nigra in Parkinson disease. ACh = acetylcholine; DA = dopamine; GABA = γ -aminobutyric acid.

of both areas. In Parkinson disease, destruction of cells in the substantia nigra results in the degeneration of the nerve terminals that secrete dopamine in the neostriatum. Thus, the normal inhibitory influence of dopamine on cholinergic neurons in the neostriatum is significantly diminished, resulting in overproduction, or a relative overactivity, of acetylcholine by the stimulatory neurons (shown in green in Figure 8.4). This triggers a chain of abnormal signaling, resulting in loss of the control of muscle movements.

3. Secondary parkinsonism: Drugs such as the phenothiazines and *haloperidol*, whose major pharmacologic action is blockade of dopamine receptors in the brain, may produce parkinsonian symptoms (also called pseudoparkinsonism). These drugs should be used with caution in patients with Parkinson disease.

B. Strategy of treatment

In addition to an abundance of inhibitory dopaminergic neurons, the neostriatum is also rich in excitatory cholinergic neurons that oppose the action of dopamine (Figure 8.4). Many of the symptoms of parkinsonism reflect an imbalance between the excitatory cholinergic neurons and the greatly diminished number of inhibitory dopaminergic neurons. Therapy is aimed at restoring dopamine in the basal ganglia and antagonizing the excitatory effect of cholinergic neurons, thus reestablishing the correct dopamine/acetylcholine balance.

VI. DRUGS USED IN PARKINSON DISEASE

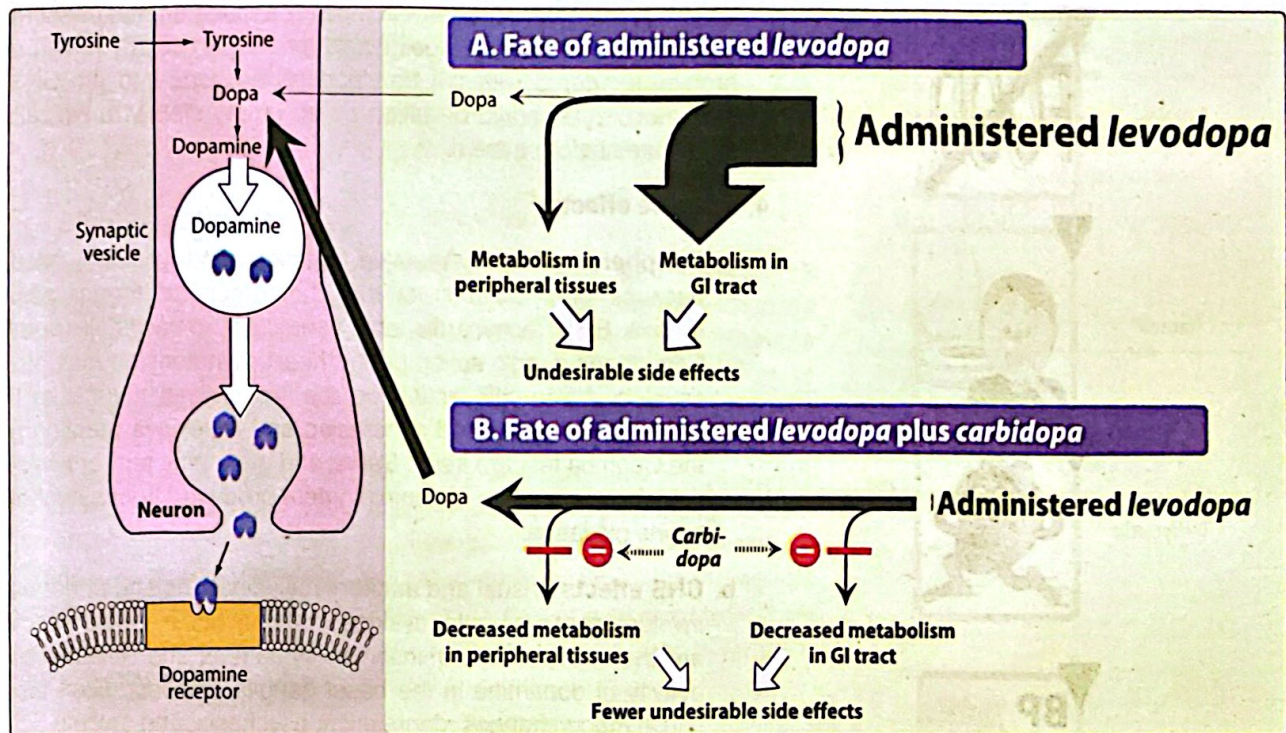
Many currently available drugs aim to maintain CNS dopamine levels, or signaling, as constant as possible. These agents offer temporary relief from the symptoms of the disorder, but they do not arrest or reverse the neuronal degeneration caused by the disease.

A. Levodopa and carbidopa

Levodopa [lee-voe-DOE-pa] is a metabolic precursor of dopamine (Figure 8.5). It restores dopaminergic neurotransmission in the neostriatum by enhancing the synthesis of dopamine in the surviving neurons of the substantia nigra. In early disease, the number of residual dopaminergic neurons in the substantia nigra (typically about 20% of normal) is adequate for conversion of *levodopa* to dopamine. Thus, in new patients, the therapeutic response to *levodopa* is consistent, and the patient rarely complains that the drug effects “wear off.” Unfortunately, with time, the number of neurons decreases, and fewer cells are capable of converting exogenously administered *levodopa* to dopamine. Consequently, motor control fluctuation develops. Relief provided by *levodopa* is only symptomatic, and it lasts only while the drug is present in the body.

1. Mechanism of action

a. Levodopa: Dopamine does not cross the blood–brain barrier, but its immediate precursor, *levodopa*, is actively transported into the CNS and converted to dopamine (Figure 8.5). *Levodopa* must be administered with *carbidopa* [kar-bi-DOE-pa]. Without

**Figure 8.5**

Synthesis of dopamine from *levodopa* in the absence and presence of *carbidopa*, an inhibitor of dopamine decarboxylase in the peripheral tissues. (GI = gastrointestinal.)

carbidopa, much of the drug is decarboxylated to dopamine in the periphery, resulting in diminished effect, nausea, vomiting, cardiac arrhythmias, and hypotension.

b. Carbidopa: *Carbidopa*, a dopamine decarboxylase inhibitor, diminishes the metabolism of *levodopa* in the periphery, thereby increasing the availability of *levodopa* to the CNS. The addition of *carbidopa* lowers the dose of *levodopa* needed by four- to five-fold and, consequently, decreases the severity of adverse effects arising from peripherally formed dopamine.

2. Therapeutic uses: *Levodopa* in combination with *carbidopa* is an efficacious drug regimen for the treatment of Parkinson disease. It decreases rigidity, tremors, and other symptoms of parkinsonism. In approximately two-thirds of patients with Parkinson disease, *levodopa-carbidopa* substantially reduces the severity of symptoms for the first few years of treatment. Patients typically experience a decline in response during the 3rd to 5th year of therapy. Withdrawal from the drug must be gradual.

3. Absorption and metabolism: The drug is absorbed rapidly from the small intestine (when empty of food). *Levodopa* has an extremely short half-life (1 to 2 hours), which causes fluctuations in plasma concentration. This may produce fluctuations in motor response, which generally correlate with the plasma concentration of *levodopa*, or perhaps give rise to the more troublesome "on-off" phenomenon, in which the motor fluctuations are not related to plasma levels in a simple way. Motor fluctuations may cause the

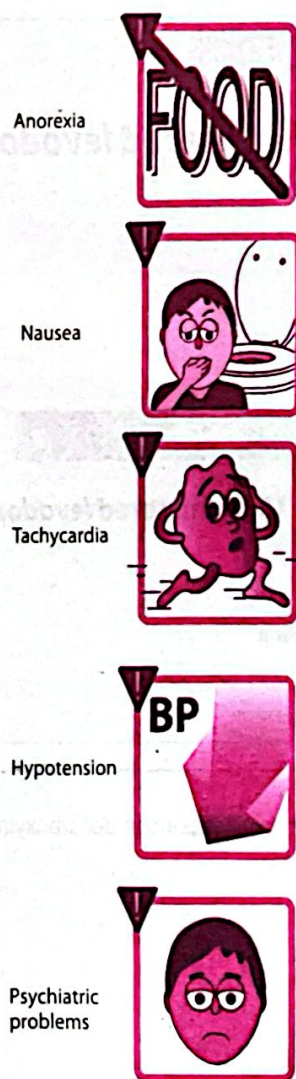


Figure 8.6
Adverse effects of *levodopa*.

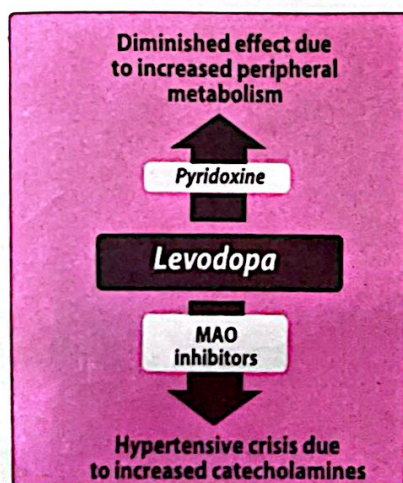


Figure 8.7
Some drug interactions observed with *levodopa*. MAO = monoamine oxidase.

patient to suddenly lose normal mobility and experience tremors, cramps, and immobility. Ingestion of meals, particularly if high in protein, interferes with the transport of *levodopa* into the CNS. Thus, *levodopa* should be taken on an empty stomach, typically 30 minutes before a meal.

4. Adverse effects

a. Peripheral effects: Anorexia, nausea, and vomiting occur because of stimulation of the chemoreceptor trigger zone (Figure 8.6). Tachycardia and ventricular extrasystole result from dopaminergic action on the heart. Hypotension may also develop. Adrenergic action on the iris causes mydriasis. In some individuals, blood dyscrasias and a positive reaction to the Coombs test are seen. Saliva and urine may turn brownish color because of the melanin pigment produced from catecholamine oxidation.

b. CNS effects: Visual and auditory hallucinations and abnormal involuntary movements (dyskinesias) may occur. These effects are the opposite of parkinsonian symptoms and reflect overactivity of dopamine in the basal ganglia. *Levodopa* can also cause mood changes, depression, psychosis, and anxiety.

5. Interactions: The vitamin pyridoxine (B_6) increases the peripheral breakdown of *levodopa* and diminishes its effectiveness (Figure 8.7). Concomitant administration of *levodopa* and non-selective monoamine oxidase inhibitors (MAOIs), such as *phenelzine*, can produce a hypertensive crisis caused by enhanced catecholamine production. Therefore, concomitant administration of these agents is contraindicated. In many psychotic patients, *levodopa* exacerbates symptoms, possibly through the buildup of central catecholamines. Cardiac patients should be carefully monitored for the possible development of arrhythmias. Antipsychotic drugs are generally contraindicated in Parkinson disease, because they potentially block dopamine receptors and may augment parkinsonian symptoms. However, low doses of atypical antipsychotics, such as *quetiapine* or *clozapine*, are sometimes used to treat *levodopa*-induced psychotic symptoms.

B. Selegiline, rasagiline, and safinamide

Selegiline [seh-LEDGE-ah-leen], also called *deprenyl* [DE-pre-nill], selectively inhibits monoamine oxidase (MAO) type B, the enzyme that metabolizes dopamine. It does not inhibit MAO type A (metabolizes norepinephrine and serotonin) unless given above recommended doses, where it loses its selectivity. By decreasing the metabolism of dopamine, *selegiline* increases dopamine levels in the brain (Figure 8.8). When *selegiline* is administered with *levodopa*, it enhances the actions of *levodopa* and substantially reduces the required dose. Unlike nonselective MAOIs, *selegiline* at recommended doses has little potential for causing hypertensive crises. However, the drug loses selectivity at high doses, and there is a risk for severe hypertension. *Selegiline* is metabolized to *methamphetamine* and *amphetamine*, whose stimulating properties may produce insomnia if the drug is administered later than mid-afternoon.

Rasagiline [ra-SA-gi-leen], an irreversible and selective inhibitor of brain MAO type B, has five times the potency of *selegiline*. Unlike *selegiline*, *rasagiline* is not metabolized to an amphetamine-like substance. *Safinamide* [sa-FIN-a-mide] is also a selective inhibitor of MAO type B indicated for use as an adjunct to *levodopa-carbidopa*.

C. Catechol-O-methyltransferase inhibitors

Normally, the methylation of *levodopa* by catechol-O-methyltransferase (COMT) to 3-O-methyldopa is a minor pathway for *levodopa* metabolism. However, when peripheral dopamine decarboxylase activity is inhibited by *carbidopa*, a significant concentration of 3-O-methyldopa is formed that competes with *levodopa* for active transport into the CNS (Figure 8.9). *Entacapone* [en-TAK-a-pone] and *tolcapone* [TOLE-ka-pone] selectively and reversibly inhibit COMT. Inhibition of COMT by these agents leads to decreased plasma concentrations of 3-O-methyldopa, increased central uptake of *levodopa*, and greater concentrations of brain dopamine. Both of these agents reduce the symptoms of "wearing-off" phenomena seen in patients on *levodopa-carbidopa*. The two drugs differ primarily in their pharmacokinetic and adverse effect profiles.

- 1. Pharmacokinetics:** Oral absorption of both drugs occurs readily and is not influenced by food. They are extensively bound to plasma albumin, with a limited volume of distribution. *Tolcapone* has a relatively long duration of action (probably due to its affinity for the enzyme) compared to *entacapone*, which requires more frequent dosing. Both drugs are extensively metabolized and eliminated in feces and urine. The dosage may need to be adjusted in patients with moderate or severe cirrhosis.
- 2. Adverse effects:** Both drugs exhibit adverse effects that are observed in patients taking *levodopa-carbidopa*, including diarrhea, postural hypotension, nausea, anorexia, dyskinesias, hallucinations, and sleep disorders. Most seriously, fulminating hepatic necrosis is associated with *tolcapone* use. Therefore, it should be used, along

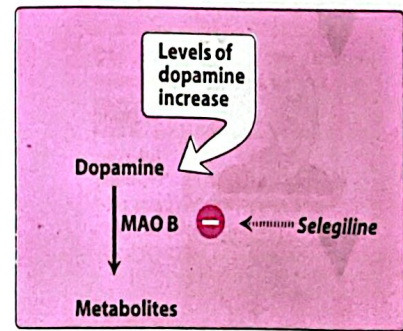


Figure 8.8

Action of *selegiline* (*deprenyl*) in dopamine metabolism. (MAO B = monoamine oxidase type B.)

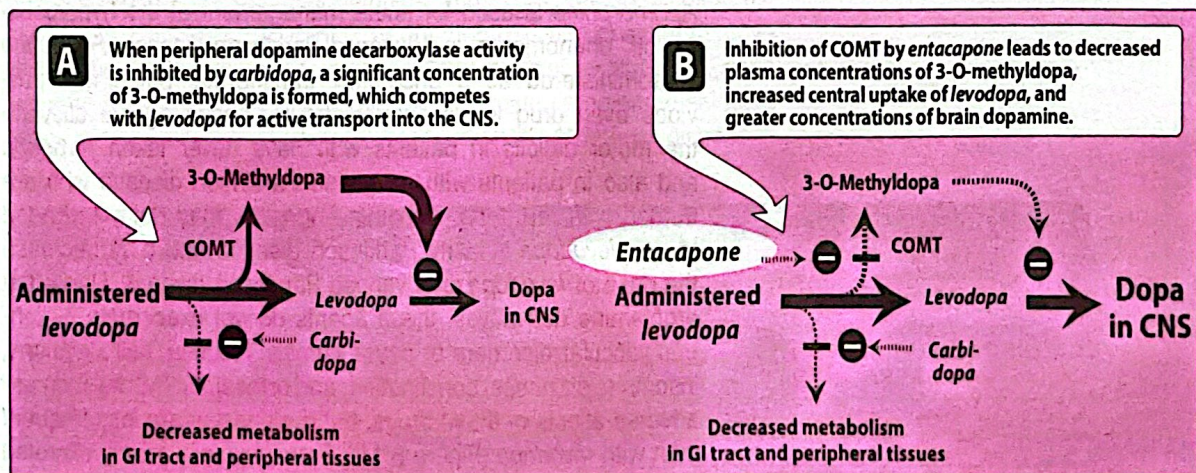
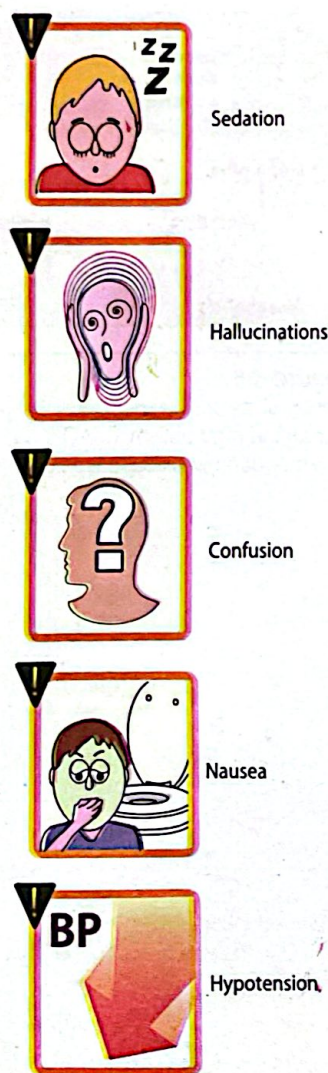


Figure 8.9

Effect of *entacapone* on dopa concentration in the central nervous system (CNS). COMT = catechol-O-methyltransferase.

**Figure 8.10**

Some adverse effects of dopamine agonists.

with appropriate hepatic function monitoring, only in patients in whom other modalities have failed. *Entacapone* does not exhibit this toxicity and has largely replaced *tolcapone* in clinical practice.

D. Dopamine receptor agonists

This group of antiparkinsonian compounds includes *bromocriptine* [broe-moe-KRIP-teen], an ergot derivative, and the nonergot drugs, *ropinirole* [roe-PIN-i-role], *pramipexole* [pra-mi-PEX-ole], *rotigotine* [ro-TIG-oh-teen], and *apomorphine* [A-poe-more-feen]. These agents have a longer duration of action than that of *levodopa* and are effective in patients exhibiting fluctuations in response to *levodopa*. Initial therapy with these drugs is associated with less risk of developing dyskinesias and motor fluctuations as compared to patients started on *levodopa*. *Bromocriptine*, *pramipexole*, and *ropinirole* are effective in patients with Parkinson disease complicated by motor fluctuations and dyskinesias. However, these drugs are ineffective in patients who have not responded to *levodopa*. *Apomorphine* is an injectable dopamine agonist that is used in severe and advanced stages of the disease to supplement oral medications. Adverse effects severely limit the utility of the dopamine agonists (Figure 8.10).

1. Bromocriptine: The actions of the ergot derivative *bromocriptine* are similar to those of *levodopa*, except that hallucinations, confusion, delirium, nausea, and orthostatic hypotension are more common, whereas dyskinesia is less prominent. In psychiatric illness, *bromocriptine* may cause the mental condition to worsen. It should be used with caution in patients with a history of myocardial infarction or peripheral vascular disease due to the risk of vasospasm. Because *bromocriptine* is an ergot derivative, it has the potential to cause pulmonary and retroperitoneal fibrosis.

2. Apomorphine, pramipexole, ropinirole, and rotigotine: These are nonergot dopamine agonists that are approved for the treatment of Parkinson disease. *Ropinirole* is also indicated for the treatment of restless legs syndrome. *Pramipexole* and *ropinirole* are orally active agents. *Apomorphine* and *rotigotine* are available in injectable and transdermal delivery systems, respectively. *Apomorphine* is used for acute management of the hypomobility "off" phenomenon in advanced Parkinson disease. *Rotigotine* is administered as a once-daily transdermal patch that provides even drug levels over 24 hours. These agents alleviate the motor deficits in patients who have never taken *levodopa* and also in patients with advanced Parkinson disease who are treated with *levodopa*. Dopamine agonists may delay the need to use *levodopa* in early Parkinson disease and may decrease the dose of *levodopa* in advanced Parkinson disease. Unlike the ergotamine derivatives, these agents do not exacerbate peripheral vascular disorders or cause fibrosis. Nausea, hallucinations, insomnia, dizziness, constipation, and orthostatic hypotension are adverse effects of these drugs, but dyskinesias are less frequent than with *levodopa* (Figure 8.11). *Pramipexole* is mainly excreted unchanged in the urine, and dosage adjustments are needed in renal dysfunction. The fluoroquinolone antibiotics and other inhibitors of the cytochrome P450 (CYP450) 1A2 isoenzyme (for example, *fluvoxamine*) may inhibit the metabolism of *ropinirole*,

requiring an adjustment in *ropinirole* dosage. Figure 8.12 summarizes some properties of dopamine agonists.

E. Amantadine

It was accidentally discovered that the antiviral drug *amantadine* [a-MAN-ta-deen] has an antiparkinsonian action. *Amantadine* has several effects on a number of neurotransmitters implicated in parkinsonism, including increasing the release of dopamine, blocking cholinergic receptors, and inhibiting the *N*-methyl-D-aspartate (NMDA) type of glutamate receptors. The drug may cause restlessness, agitation, confusion, and hallucinations, and, at high doses, it may induce acute toxic psychosis. Orthostatic hypotension, urinary retention, peripheral edema, and dry mouth also may occur. *Amantadine* is less efficacious than *levodopa*, and tolerance develops more readily. However, *amantadine* has fewer adverse effects.

F. Antimuscarinic agents

The antimuscarinic agents are much less efficacious than *levodopa* and play only an adjuvant role in antiparkinsonism therapy. The actions of *benztropine* [BENZ-troe-peen] and *trihexyphenidyl* [tri-hex-ee-FEN-i-dill] are similar, although individual patients may respond more favorably to one drug or the other. Blockade of cholinergic transmission produces effects similar to augmentation of dopaminergic transmission, since it helps to correct the imbalance in the dopamine/acetylcholine activity (Figure 8.4). These agents can induce mood changes and confusion, and produce xerostomia, constipation, and visual problems typical of muscarinic blockers (see Chapter 5). They interfere with gastrointestinal peristalsis and are contraindicated in patients with glaucoma, prostatic hyperplasia, or pyloric stenosis.

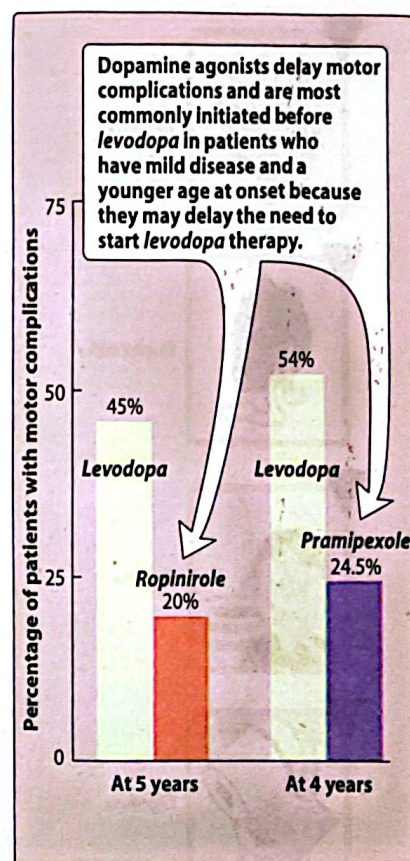


Figure 8.11
Motor complications in patients treated with *levodopa* or dopamine agonists.

VII. DRUGS USED IN ALZHEIMER DISEASE

Dementia of the Alzheimer type has three distinguishing features: 1) accumulation of senile plaques (β -amyloid accumulations), 2) formation of numerous neurofibrillary tangles, and 3) loss of cortical neurons, particularly cholinergic neurons. Current therapies aim to either improve cholinergic transmission within the CNS or prevent excitotoxic actions resulting from overstimulation of NMDA-glutamate receptors in selected areas of the brain. Pharmacologic intervention for Alzheimer disease is

Characteristic	<i>Pramipexole</i>	<i>Ropinirole</i>	<i>Rotigotine</i>
Bioavailability	>90%	55%	45%
V_d	7 L/kg	7.5 L/kg	84 L/kg
Half-life	8 h ¹	6 h	7 h ³
Metabolism	Negligible	Extensive	Extensive
Elimination	Renal	Renal ²	Renal ²

Figure 8.12

Pharmacokinetic properties of dopamine agonists *pramipexole*, *ropinirole*, and *rotigotine*. V_d = volume of distribution. ¹Increases to 12 hours in patients older than 65 years. ²Less than 10% excreted unchanged.

³Administered as a once-daily transdermal patch.